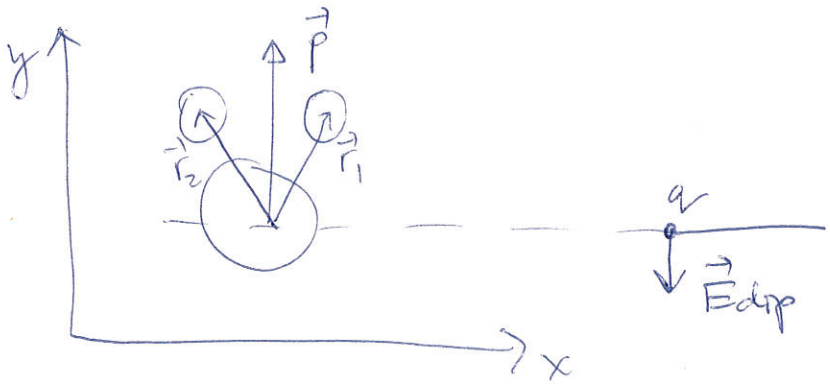
Midterm 2

P1

$$\rho(\vec{r}) = q\left(\delta(\vec{r}-\vec{r}_1) + \delta(\vec{r}-\vec{r}_2) - 2\delta(\vec{r})\right)$$

$$\vec{p} = \int \vec{r}\, \rho(\vec{r})\, d^3\vec{r} = q\left(\vec{r}_1 + \vec{r}_2 - \vec{0}\right) = p\,\hat{y}$$



Electric field at point $\vec{r} = a\hat{x}$

$$\vec{E}_{dip}(\vec{x}) = \frac{1}{4\pi\varepsilon_0}\,\frac{3(\vec{p}\cdot\hat{x})\hat{x} - \vec{p}}{a^3}$$

but $\vec{p}\cdot\hat{x} = p\,\hat{y}\cdot\hat{x} = 0 \Rightarrow \vec{E}_{dip}(x) = \dfrac{-\vec{p}}{4\pi\varepsilon_0 a^3}$

$\Rightarrow$ force on charge q is point $\vec{r} = a\hat{x}$ is

$$\vec{F} = \frac{-p}{4\pi\varepsilon_0 a^3}\,\hat{y}$$ is along $-\hat{y}$ axis

$$\cancel{W = \frac{1}{4\pi\varepsilon_0}\frac{Q^2}{2R} = \frac{(4\pi\sigma)^2}{8\pi\varepsilon_0}R}$$

$$\cancel{Q = 4\pi R\sigma}$$

$$\cancel{W_{surface} = \lambda\, 4\pi R^2}$$

P2

$$\rho(r) = \begin{cases} br & \text{for } R_1 \le r \le R_2 \\ 0 & \text{otherwise} \end{cases} \quad \cancel{\rho(r) = \lambda r}$$

$$\int \vec{E}\cdot d\vec{S} = \frac{1}{\varepsilon_0}\underbrace{\int \rho(r)\,d^3r}_{\text{enclosed charge}}$$

$$\Rightarrow \vec{E} = 0 \text{ for } r < R_1$$

$$E\cdot 4\pi r^2 = \frac{1}{\varepsilon_0}\int br\cdot 4\pi r^2 dr = \frac{4\pi}{\varepsilon_0} b\int_{R_1}^{r} r^3 dr =$$

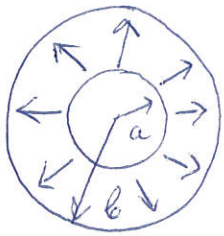
$$= \frac{4\pi b}{\varepsilon_0}\,\frac{1}{4}\left(r^4 - R_1^4\right)$$

$$\vec{E} = \frac{b}{4\varepsilon_0}\,\frac{r^4 - R_1^4}{r^2}\,\hat{r} \text{ for } R_1 < r < R_2$$

$$E\cdot 4\pi r^2 = \frac{4\pi b}{\varepsilon_0}\int_{R_1}^{R_2} r^3 dr = \frac{\pi b}{\varepsilon_0}\left(R_2^4 - R_1^4\right)$$

$$\vec{E} = \frac{\pi b}{4\pi\varepsilon_0}\,\frac{R_2^4 - R_1^4}{r^2}\,\hat{r} \text{ for } r > R_2$$

P3



$$\frac{1}{s}\frac{\partial}{\partial s}\left(s\frac{\partial V}{\partial s}\right)+\frac{1}{s^2}\frac{\partial^2 V}{\partial \varphi^2}+\frac{\partial^2 V}{\partial z^2}=0$$

but $V=V(s)$ only

$$\Rightarrow \frac{1}{s}\frac{d}{ds}\left(s\frac{dV}{ds}\right)=0$$

$$\Rightarrow \quad s\frac{dV}{ds}=\text{const}=c$$

$$\frac{dV}{ds}=\frac{c}{s}$$

$$\Rightarrow V(s)=A+B\ln\left(\frac{s}{s_0}\right)$$

$$\left.\begin{aligned} V(s=a)&=V_a=A+B\ln\left(\frac{a}{s_0}\right)\\ V(s=b)&=V_b=A+B\ln\left(\frac{b}{s_0}\right)\end{aligned}\right\}\quad \begin{aligned} V_a-V_b&=B\ln\left(\frac{a}{b}\right)\\ B&=\frac{V_a-V_b}{\ln\left(\frac{a}{b}\right)}.\end{aligned}$$

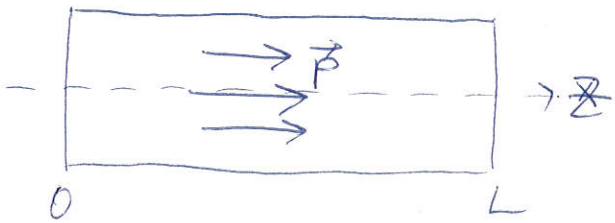
$$\vec{E}=-\vec{\nabla}V=-\left(\hat{s}\frac{\partial V}{\partial s}+\frac{1}{s}\frac{\partial V}{\partial \varphi}\hat{\varphi}+\frac{\partial V}{\partial z}\hat{z}\right)=$$

$$=-\frac{dV}{ds}\hat{s}=-\frac{B}{s}\hat{s}$$

$$\vec{E}=\frac{V_a-V_b}{\ln\left(\frac{b}{a}\right)}\frac{1}{s}\hat{s}$$

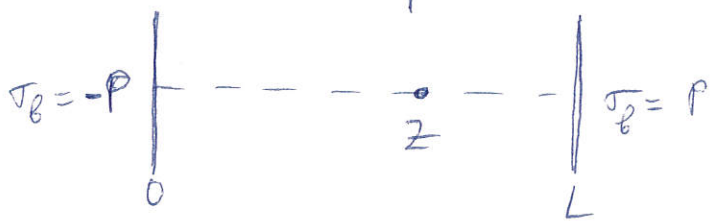
↳ radial direction

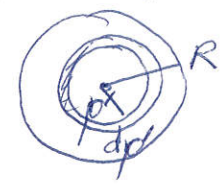
P4



$\vec{P} = \text{const} = P\hat{z}$

Equivalent problem: two oppositely charged faces of the cylinder.





$$V(z) = \frac{1}{4\pi\varepsilon_0}\int \frac{\sigma(r')}{|\vec{r}-\vec{r}'|}\,da' = V_- + V_+$$

$$\vec{r} = z\hat{z}, \quad \vec{r}' = x'\hat{x} + y'\hat{y}$$

$$V_- = \frac{-P}{4\pi\varepsilon_0}\int \frac{dx'dy'}{|z\hat{z} - x'\hat{x} - y'\hat{y}|}$$

$$V_+ = \frac{P}{4\pi\varepsilon_0}\int \frac{dx'dy'}{|(L-z)\hat{z} - x'\hat{x} - y'\hat{y}|}$$

$$V(z) = \frac{-P}{4\pi\varepsilon_0}\int \frac{dx'dy'}{\sqrt{z^2+(x')^2+(y')^2}} = \frac{-P}{4\pi\varepsilon_0}\int_0^R \frac{2\pi\rho\,d\rho}{\sqrt{z^2+\rho^2}}$$

$$= -\frac{P}{4\varepsilon_0}\int \frac{d(\rho^2)}{\sqrt{z^2+\rho^2}} = -\frac{P}{4\varepsilon_0}\int_0^R 2\left(d\sqrt{z^2+\rho^2}\right) =$$

$$= -\frac{P}{2\varepsilon_0}\left(\sqrt{z^2+R^2} - |z|\right);$$

To get V_+, we only need to replace z by $L-z$

$$V_+ = \frac{P}{2\varepsilon_0}\left(\sqrt{(L-z)^2+R^2} - |L-z|\right)$$

$$V(z) = \frac{P}{2\varepsilon_0}\left(\sqrt{(L-z)^2+R^2} - (L-z) - \sqrt{z^2+R^2} + z\right)$$

for $0 < z < L$

$\vec{E} = -\vec{\nabla} V$. On the z-axis, by symmetry $E_x = E_y = 0$

$\Rightarrow \frac{\partial V}{\partial x} = \frac{\partial V}{\partial y} = 0$ too,

then $$\vec{E}(z) = E_z(z) = -\frac{\partial V}{\partial z} = -\frac{P}{2\varepsilon_0}\left\{2 + \frac{(L-z)(-1)}{\sqrt{(L-z)^2+R^2}} - \frac{z}{\sqrt{z^2+R^2}}\right\} = \frac{P}{2\varepsilon_0}\left\{\frac{z}{\sqrt{z^2+R^2}} + \frac{L-z}{\sqrt{(L-z)^2+R^2}} - 2\right\}$$

Limit $L \ll R$, then $z \in (0,L) \ll R$ as well,

thus $$\frac{z}{\sqrt{z^2+R^2}} \approx \frac{z}{R} \ll 1$$

$$\frac{L-z}{\sqrt{(L-z)^2+R^2}} \approx \frac{L-z}{R} \ll 1$$

$$\Rightarrow \vec{E} = -\frac{P}{\varepsilon_0}\hat{z}$$

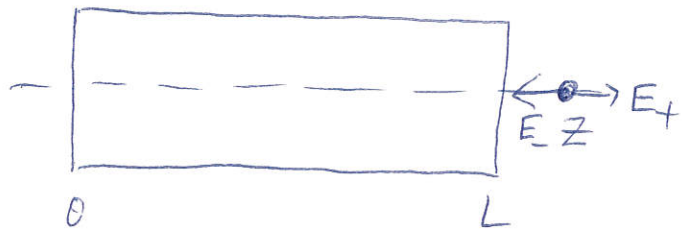
Electric displacement $\vec{D} = \varepsilon_0 \vec{E} + \vec{P} = -P\hat{z} + P\hat{z} = 0.$

Electric field outside the cylinder:

$z > L \Rightarrow |z| = z, \; |L-z| = z - L$

$$V(z) = \frac{\rho}{2\varepsilon_0}\left(z - \sqrt{z^2+R^2} + \sqrt{(L-z)^2+R^2} - (z-L)\right)$$

$$= \frac{\rho}{2\varepsilon_0}\left(L + \sqrt{(L-z)^2+R^2} - \sqrt{z^2+R^2}\right)$$



$$\vec{E} = -\frac{\partial V}{\partial z}\hat{z} = -\frac{\rho}{2\varepsilon_0}\hat{z}\left(\frac{-(L-z)}{\sqrt{(L-z)^2+R^2}} - \frac{z}{\sqrt{z^2+R^2}}\right)$$

$$= \hat{z}\,\frac{\rho}{2\varepsilon_0}\left(\frac{z}{\sqrt{z^2+R^2}} + \frac{L-z}{\sqrt{(L-z)^2+R^2}}\right)$$

$\Rightarrow$ in the $R \gg L$ limit $\vec{E} = \hat{z}\,\frac{\rho}{2\varepsilon_0}\left(\frac{z}{R} + \frac{L-z}{R}\right)$

$$= \hat{z}\,\frac{\rho}{2\varepsilon_0}\,\frac{L}{R} \to 0$$

for $\frac{L}{R} \to 0$.